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EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

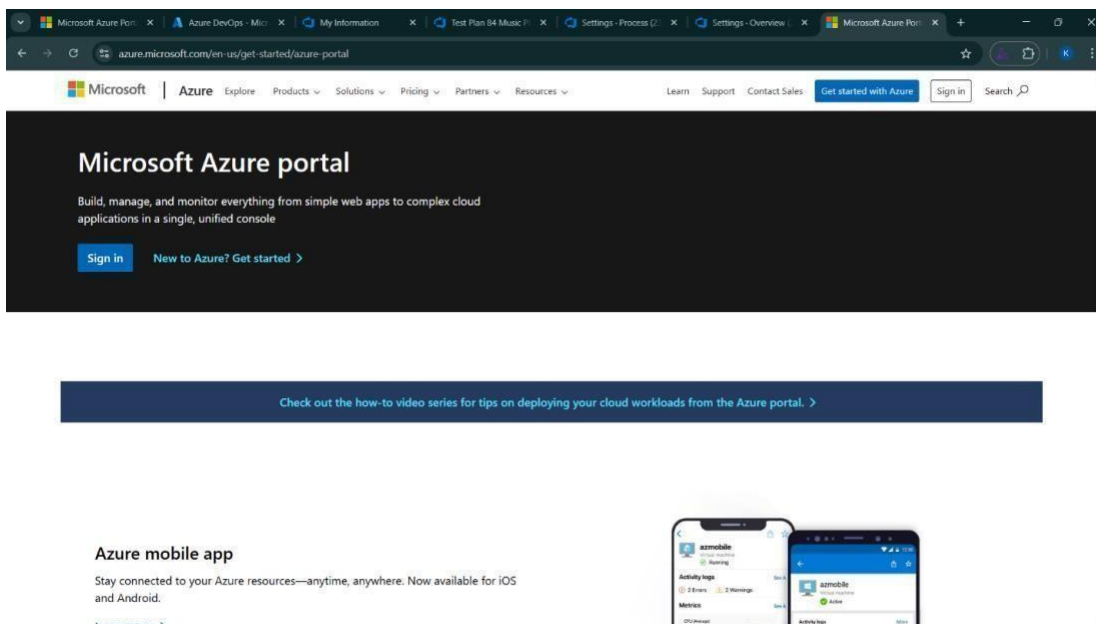
AIM

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

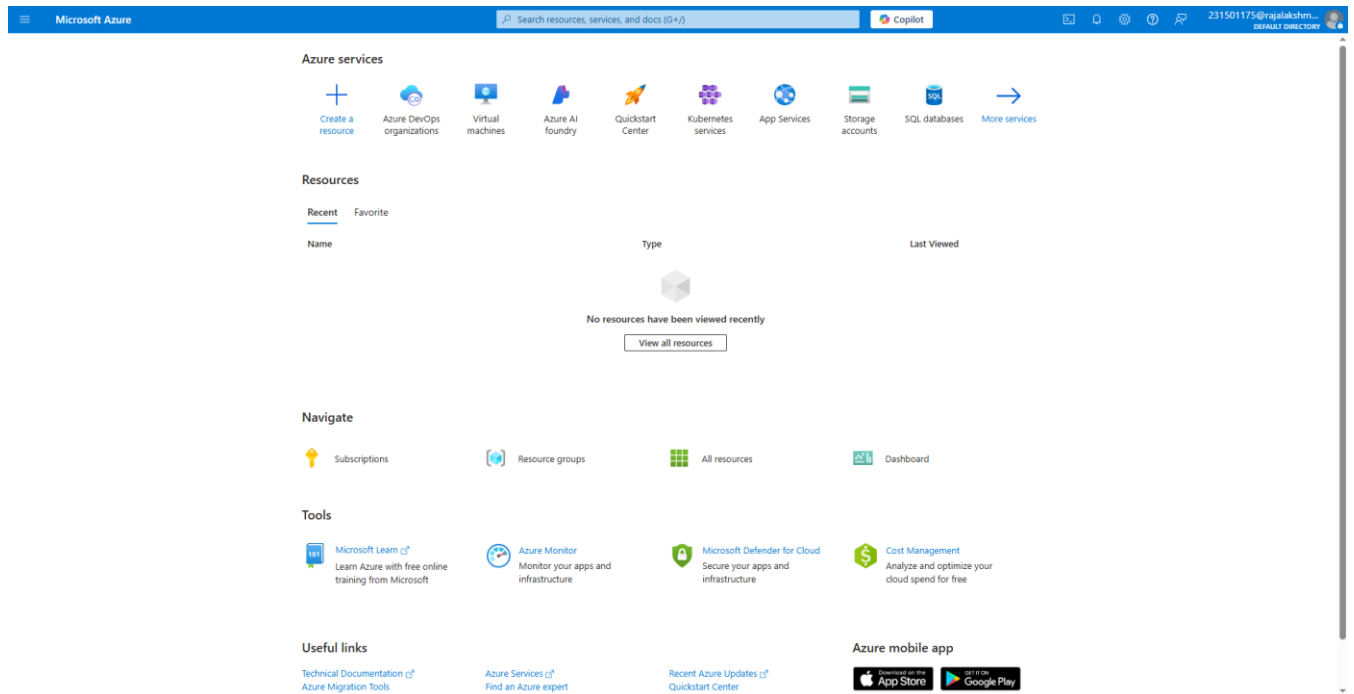
INSTALLATION

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/getstarted/azureportal>. Sign in using your Microsoft account credentials.

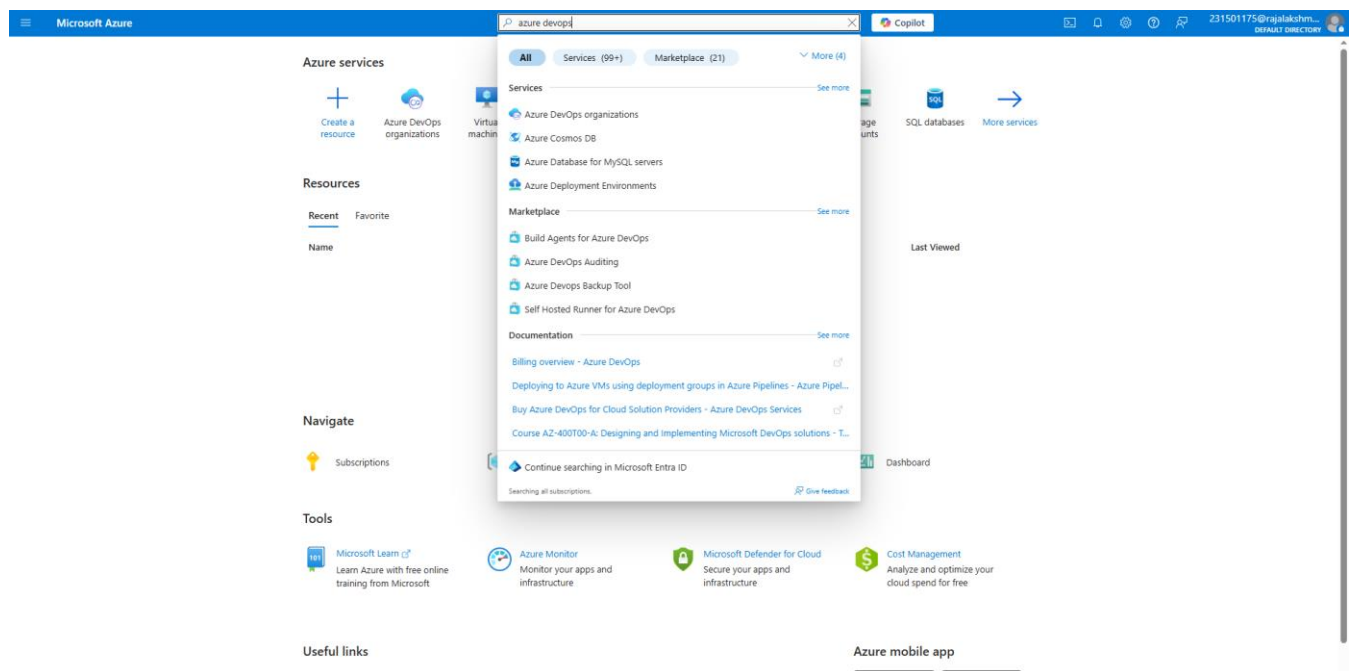
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



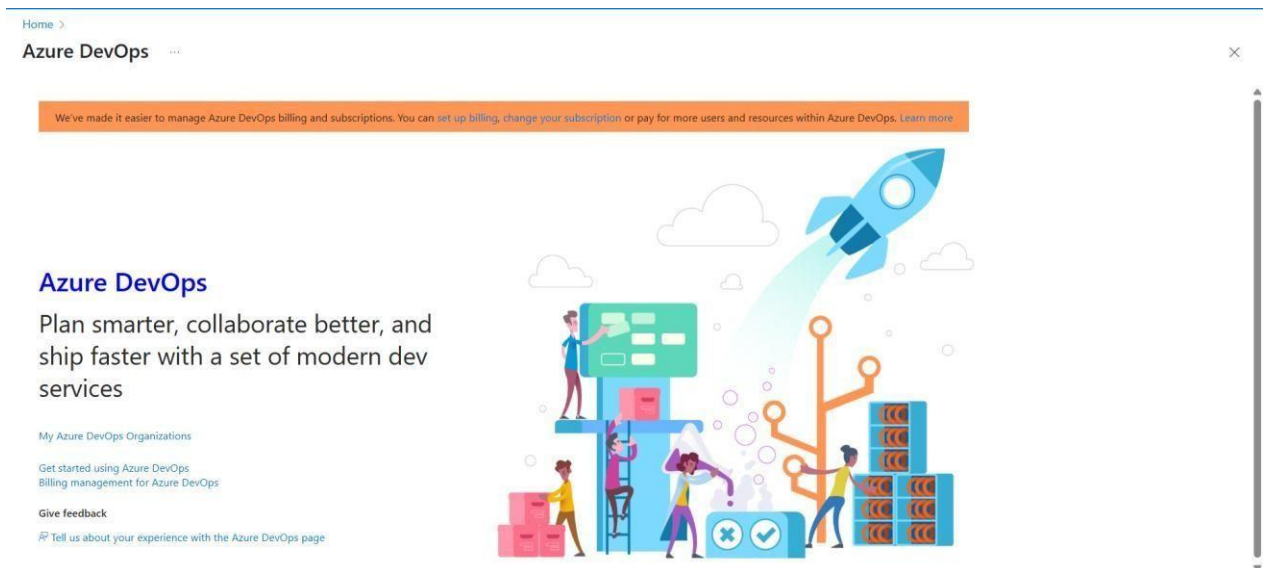
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

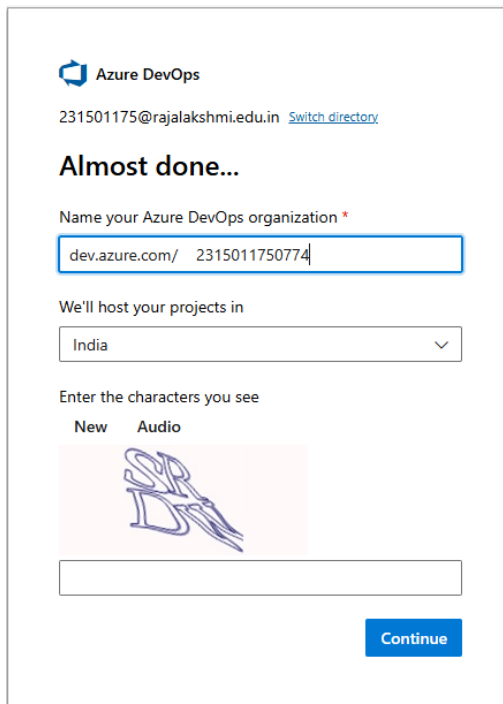
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

c. Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., **LMS**).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.

The screenshot shows the 'Create a project to get started' form. It includes a 'Project name' field with 'OQS' entered, a 'Description' field, and a 'Visibility' section with 'Public' and 'Private' options. The 'Private' option is selected. Below this is a note about public projects being disabled for the organization. There is an 'Advanced' section with 'Version control' set to 'Git' and 'Work item process' set to 'Agile'. A blue '+ Create project' button is at the bottom.

Create a project to get started

Project name *

OQS

Description

Visibility

☐ Public
Anyone on the internet can view the project. Certain features like TFVC are not supported.

☒ Private
Only people you give access to will be able to view this project.

Public projects are disabled for your organization. You can turn on public visibility with [organization policies](#).

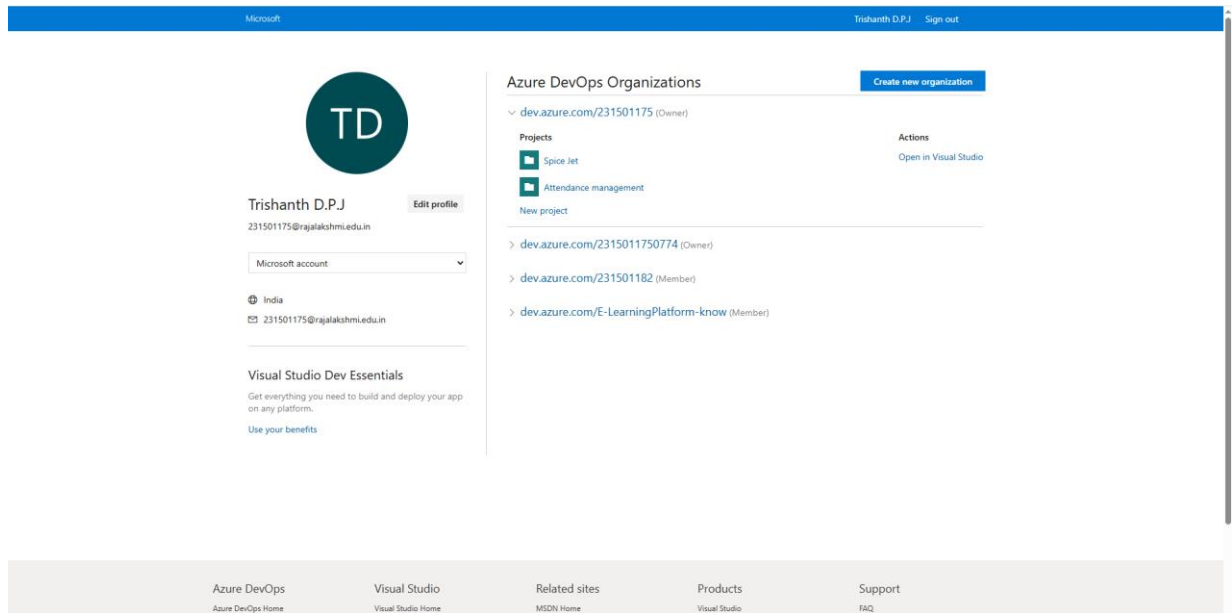
^ Advanced

Version control ⓘ
Git

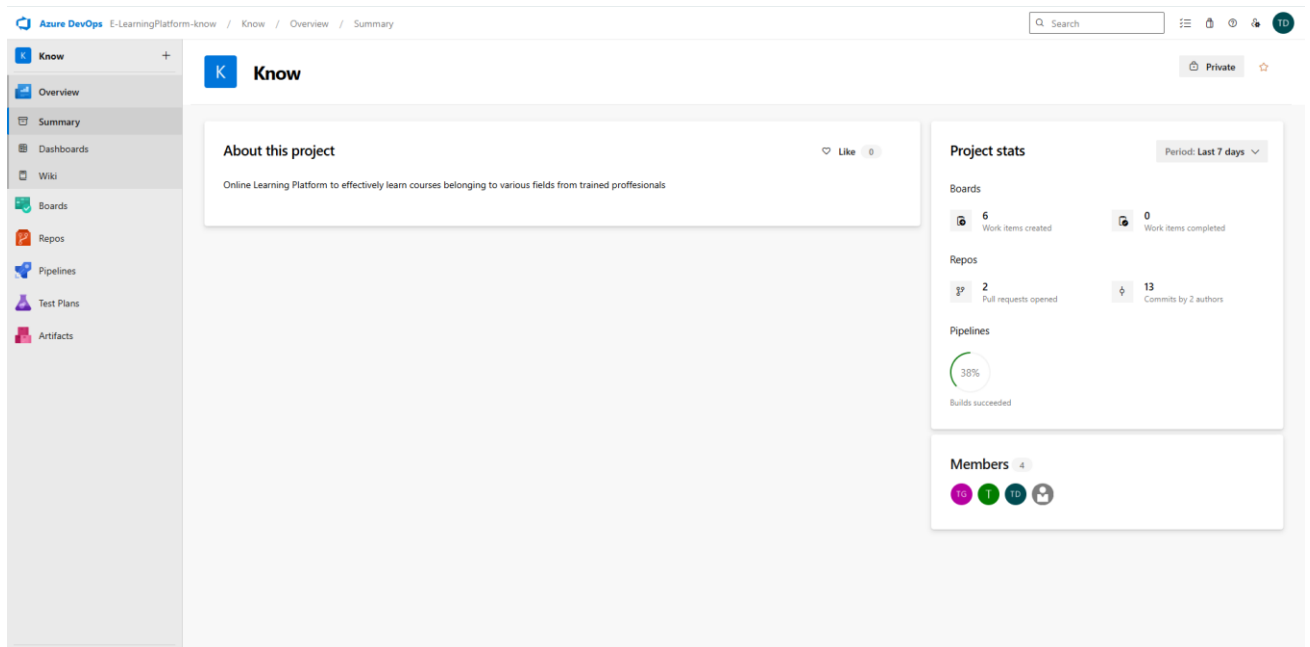
Work item process ⓘ
Agile

+ Create project

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



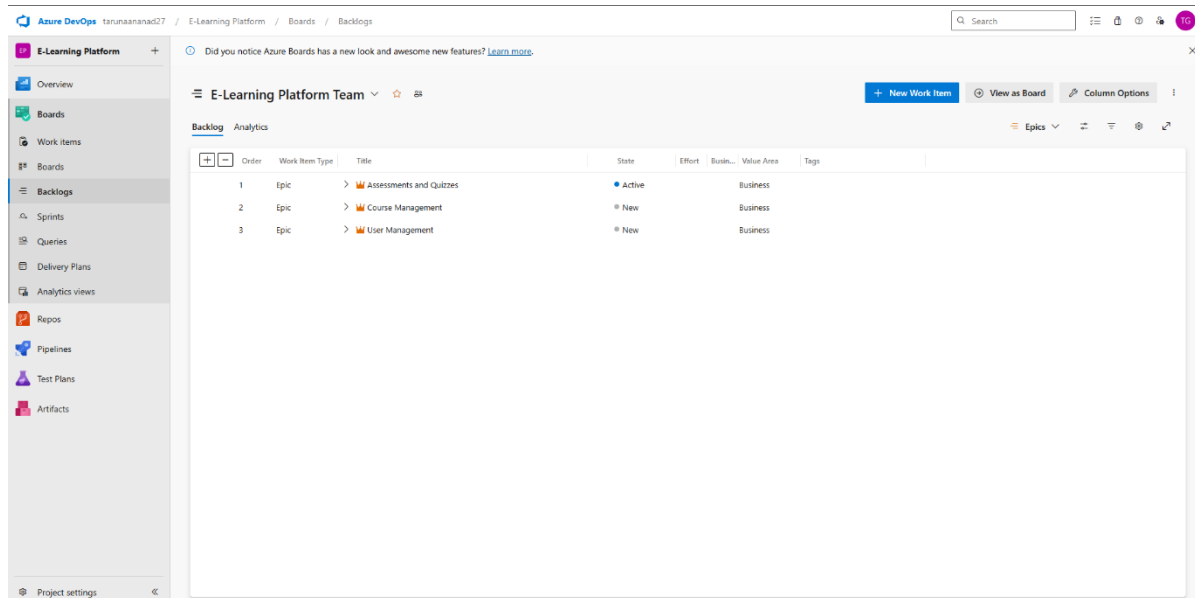
4.Project dashboard



5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



RESULT

Successfully created an Azure DevOps project with user story management and agile workflow setup.

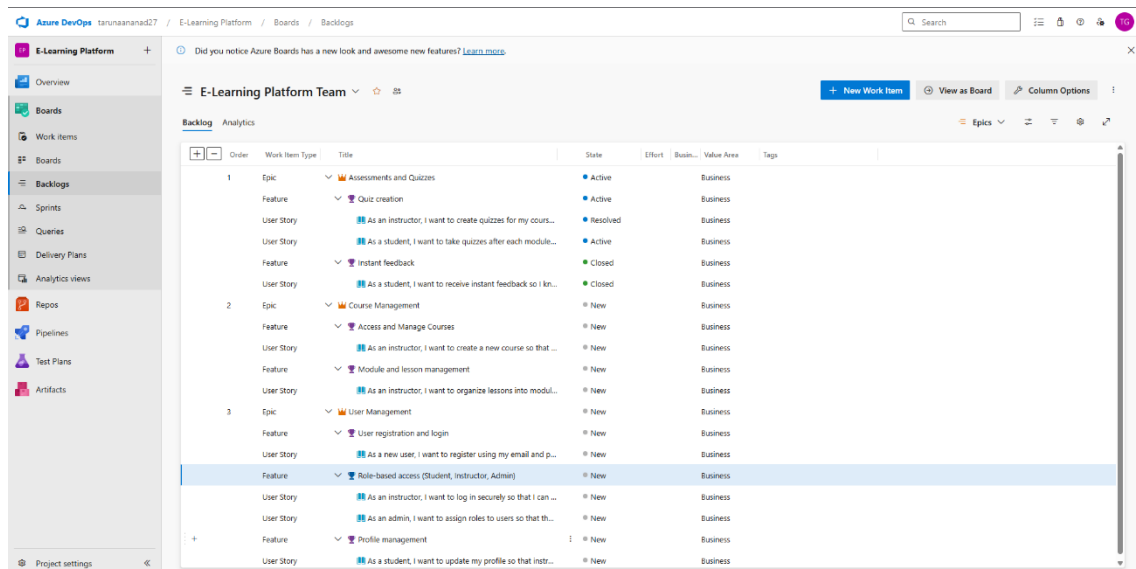
EXP NO:3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

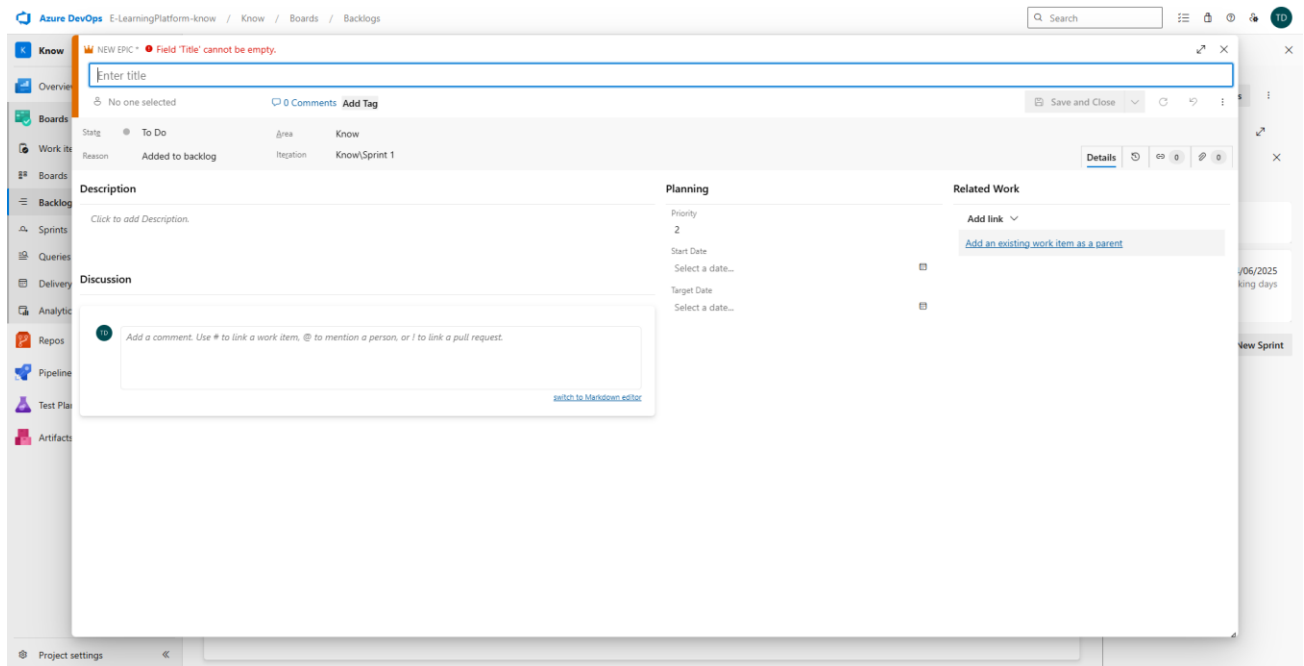
AIM

To learn about how to create epics, user story, features, backlogs for your assigned project.

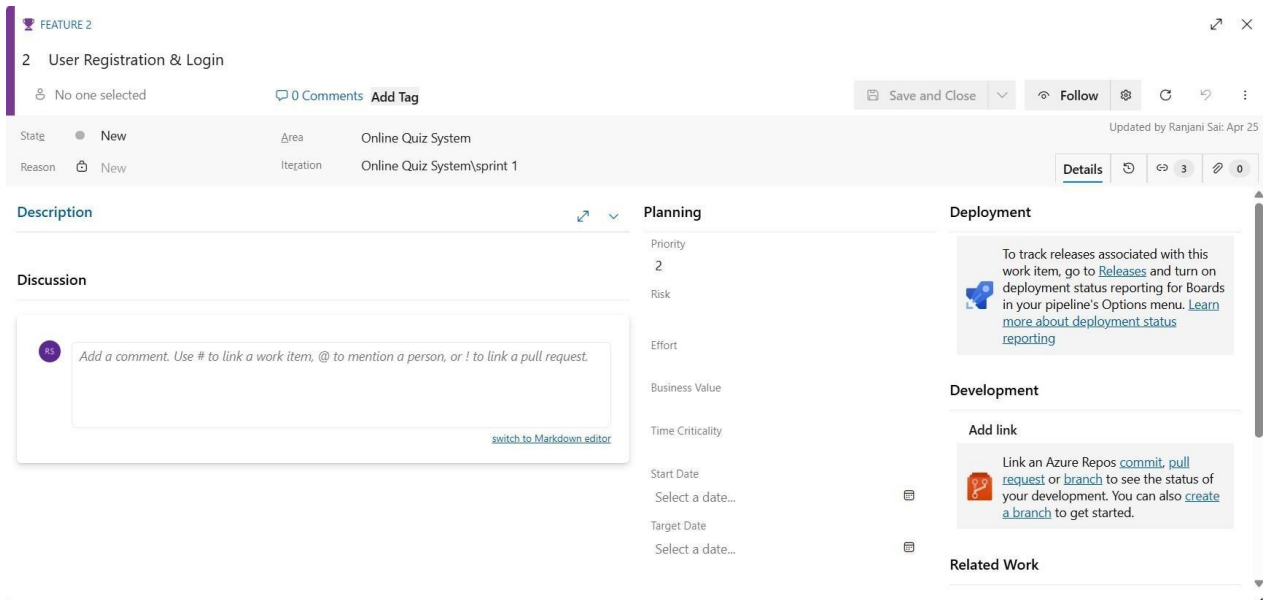
Create Epic, Features, User Stories, Task



1.Fill in Epics



2.Fill in Features



3.Fill in User Story Details

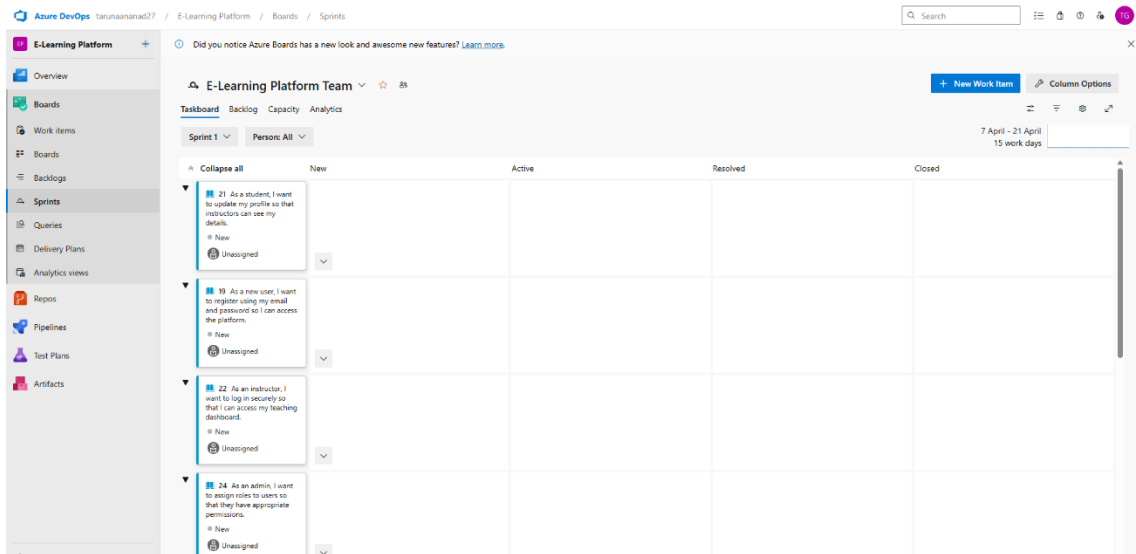
EXP NO: 4

SPRINT PLANNING

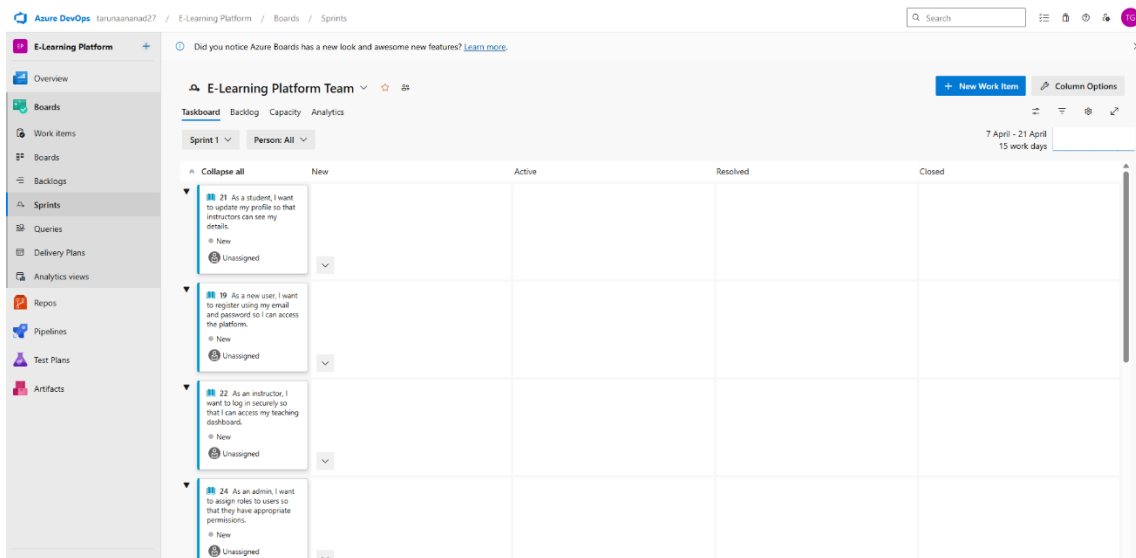
AIM

To assign user story to specific sprint for the E-Learning Platform.

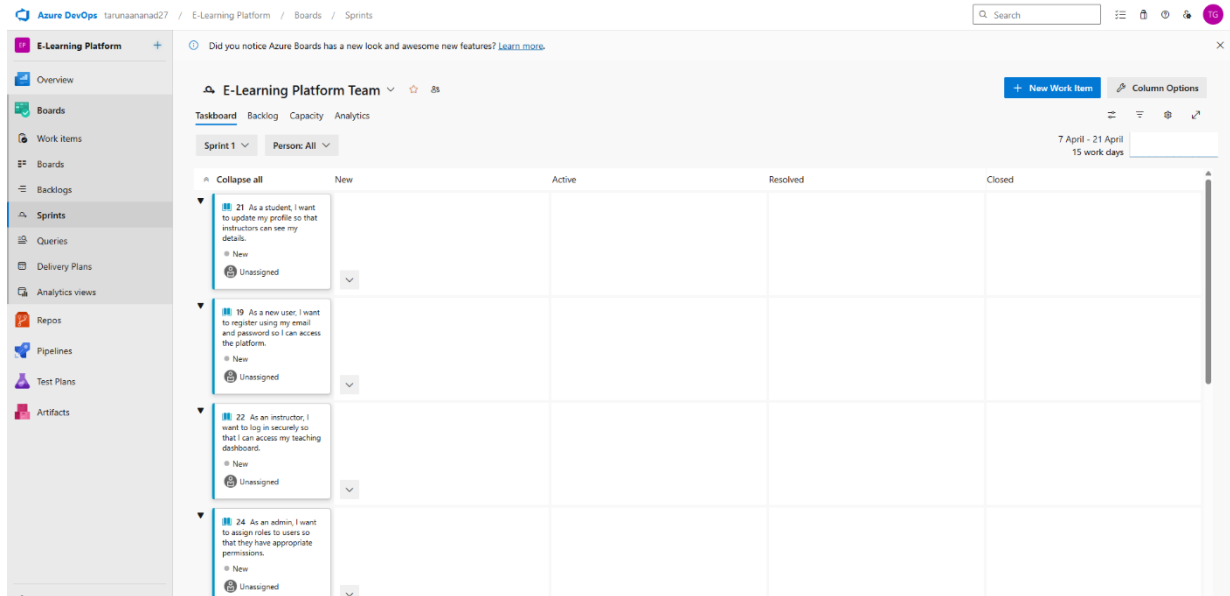
Sprint Planning Sprint 1



Sprint 2



Sprint 3



RESULT

The Sprints have been created for E-Learning Platform.

EXP NO:5

POKER ESTIMATION

AIM

Create Poker Estimation for the user stories – E-Learning Platform.

Poker Estimation

USER STORY 13

13 As an admin, I want to manage a question bank so I can reuse questions easily.

No one selected 0 Comments Add Tag

Save and Close Follow

Updated by Ranjani Sai: Apr 25

State: New Area: Online Quiz System Reason: New Iteration: Online Quiz System\sprint 1

Description

Click to add Description.

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning

Story Points

Priority 2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

RESULT

The Estimation/Story Points is created for the project using Poker Estimation.

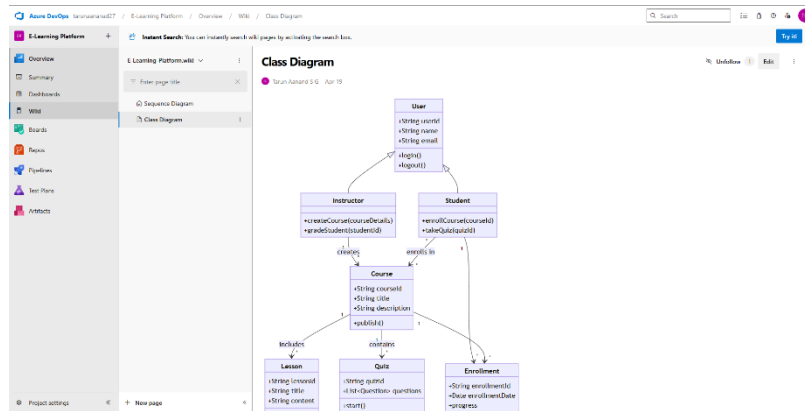
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

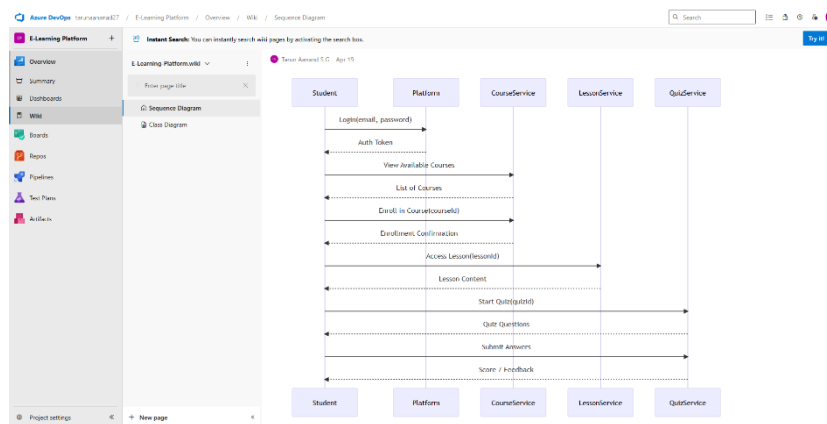
AIM

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



RESULT

The Class Diagram and Sequence Diagram is designed Successfully for the E-Learning Platform.

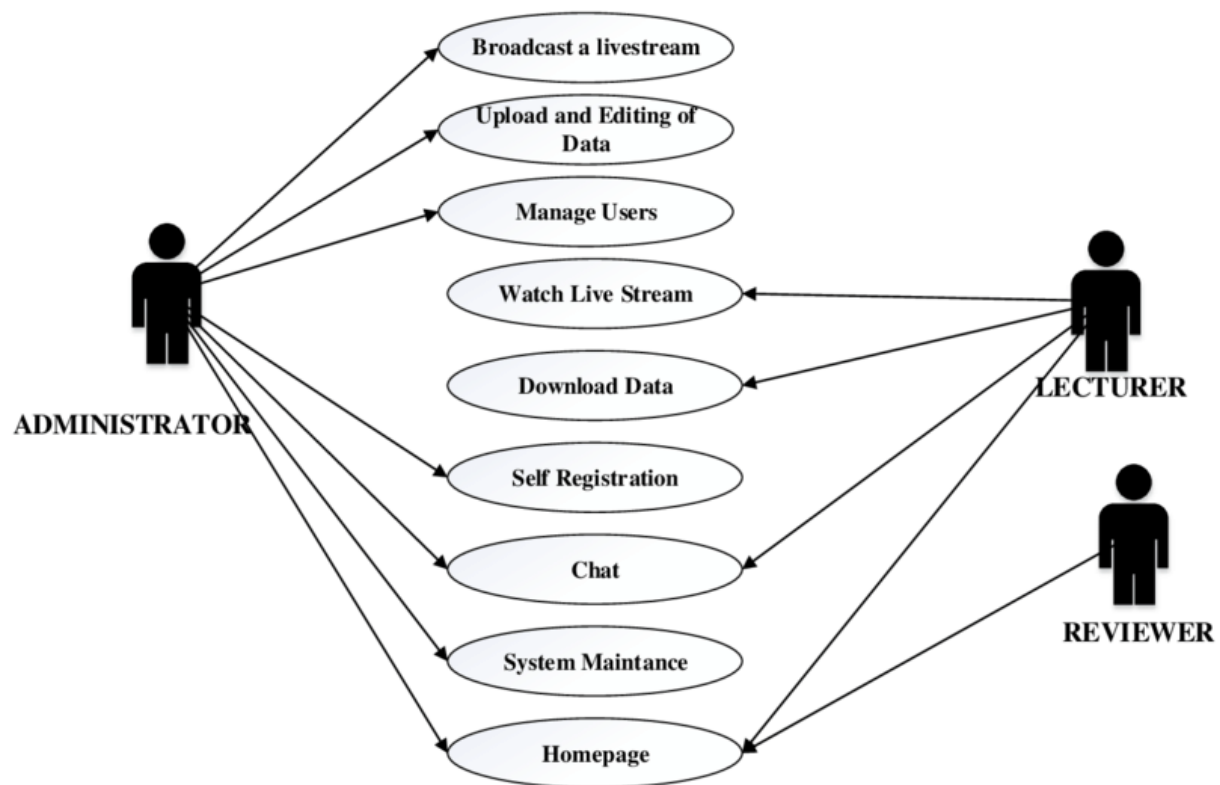
EXP NO: 7

DESIGNING USE-CASE AND ACTIVITY DIAGRAMS FOR PROJECT STRUCTURE

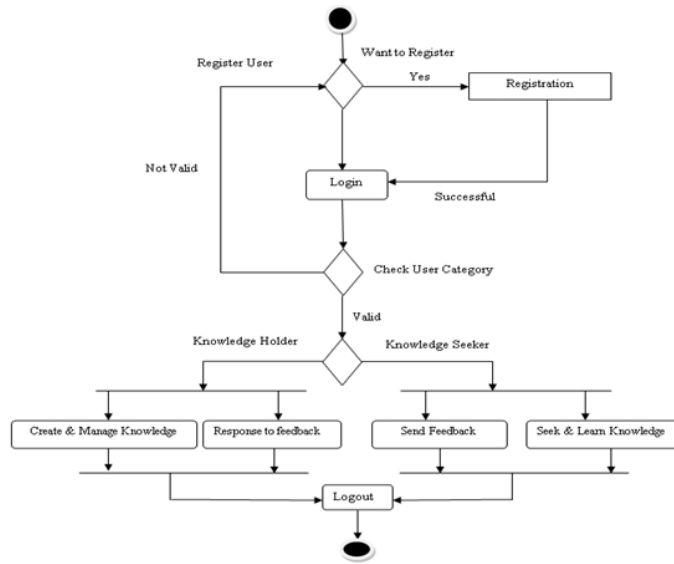
AIM

To Design an Use-Case Diagram and Activity Diagram for the given Project. 7A. Use-Case

Diagram



7B. Activity Diagram



RESULT

The Use-Case Diagram and Activity Diagram is designed Successfully for the E-Learning Platform.

EXP NO: 8

TESTING TEST PLANS AND TEST CASES

AIM

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case Test Case Design Procedure

1. Understand Core Features of the Application

- User Login

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, submitting quizzes, viewing results)

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions
- Example: Player registers and logs in, submits quizzes and views results

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues or failed actions to ensure proper error handling.
- Example: Login with invalid credentials, submission without attachments, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identifies (e.g., TC01 – Valid Login, TC03 – Invalid Password).
- Facilities easy mapping to features and tracking in Azure DevOps.

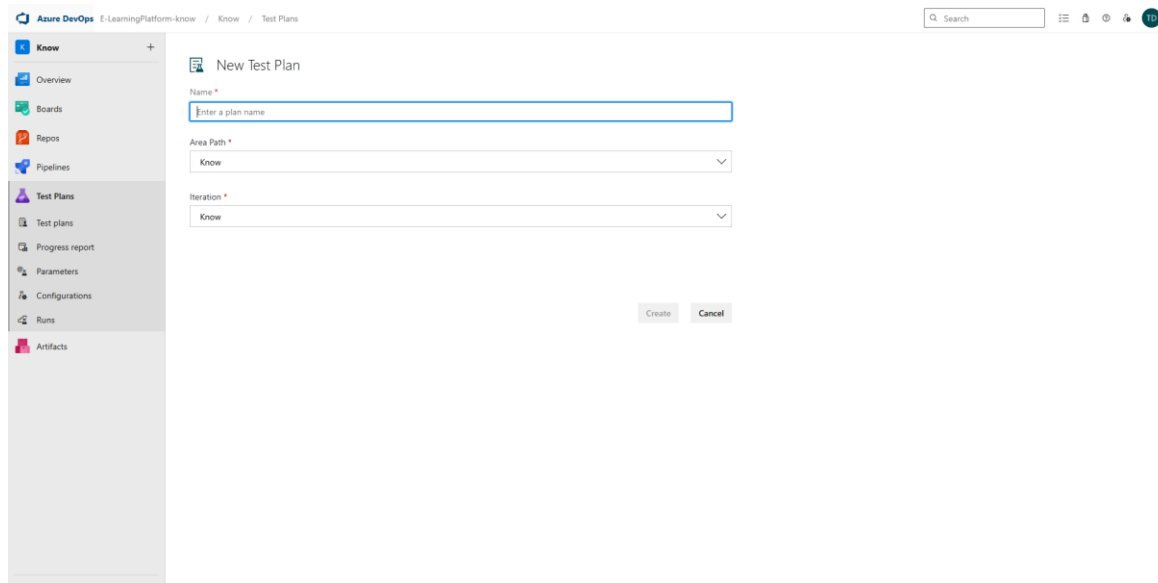
7. Separate Test

- Grouped by functionality such as:
 - Login and Registration
 - Quizzes Submission
 - Viewing Results
 - Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

8. Prioritize and Review

- High-priority assigned to critical workflows like login, quizzes and results.
- Reviewed for completeness, accuracy and alignment with user stories and features definition.

1.New test plan



The screenshot shows the 'New Test Plan' interface in Azure DevOps. The left sidebar contains a navigation menu with options: Know, Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main content area is titled 'New Test Plan' and includes three required fields: 'Name *' with a text input placeholder 'Enter a plan name', 'Area Path *' with a dropdown menu showing 'Know', and 'Iteration *' with a dropdown menu showing 'Know'. At the bottom right of the form are 'Create' and 'Cancel' buttons. The top of the interface shows the breadcrumb 'Azure DevOps E-LearningPlatform-know / Know / Test Plans' and a search bar.

2. Test suite

The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation options: Know, Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays the 'Chapter-wise Course Learning Flow (2)' test suite. It includes a 'Test Suites' section with a filter and a table of test points.

Title	Outcome	Order	Test Case Id	Configuration	Tester
Test Case 1: Open a course and view chapters	In Progress	1	8	Windows 10	Tarun Aanand S G
Test Case 2: Complete a chapter and mark as done	Failed	2	9	Windows 10	Tarun Aanand S G

3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

E-Learning Platform– Test Plans

USER STORIES

- As an admin, I want to log in to manage quizzes and users (ID: 3).
- As a player, I want to register and log in so that I can access quizzes securely (ID: 4).
- As an admin, I want to give users the right access so they can use the system properly (ID: 6).
- As a player, I want to see only my quizzes and progress so that it's easy to use (ID: 7).
- As an admin, I want to create and configure quizzes with time limits so I can control quiz flow (ID: 10).

TEST SUITES

Test Suite 1: User Registration and Login

Test Case 1.1: User Registration with Valid Details

- **Action:**
 1. Navigate to the registration page
 2. Enter valid name, email, password
 3. Click "Register"
 - **Expected Result:**

User account is created, and a confirmation message is shown.
 - **Type:** Happy Path
-

Test Case 1.2: Login with Valid Credentials

- **Action:**
 1. Go to login page
 2. Enter registered email and correct password
 3. Click "Login"
 - **Expected Result:**

User is redirected to dashboard
 - **Type:** Happy Path
-

Test Case 1.3: Logout from Application

- **Action:**
 1. Click on user profile
 2. Click "Logout"
 - **Expected Result:**

User is logged out and redirected to login page
 - **Type:** Happy Path
-

Test Suite 2: Course Enrollment and Access

Test Case 2.1: View Available Courses

- **Action:**
 1. Login to the platform
 2. Navigate to "Courses" tab
 - **Expected Result:**

A list of available courses is displayed
 - **Type:** Happy Path
-

Test Case 2.2: Enroll in a Free Course

- **Action:**
 1. From course list, select a free course
 2. Click “Enroll”
 - **Expected Result:**
User is successfully enrolled in the course
 - **Type:** Happy Path
-

Test Case 2.3: Access Enrolled Course Content

- **Action:**
 1. Go to “My Courses”
 2. Click on enrolled course
 3. Play a video lecture
 - **Expected Result:**
Course content loads and plays correctly
 - **Type:** Happy Path
-

Test Suite 3: Assessment and Certification

Test Case 3.1: Start Quiz for a Completed Module

- **Action:**
 1. Complete a module
 2. Click “Take Quiz”
 - **Expected Result:**
Quiz page opens with questions related to the module
 - **Type:** Happy Path
-

Test Case 3.2: Submit Quiz Answers

- **Action:**
 1. Select answers for all questions
 2. Click “Submit”
 - **Expected Result:**
Submission is accepted and score is displayed
 - **Type:** Happy Path
-

Test Case 3.3: Download Certificate After Course Completion

- **Action:**
 1. Complete all modules and pass final quiz
 2. Click “Download Certificate”
- **Expected Result:**
Certificate is downloaded in PDF format
- **Type:** Happy Path

Test Cases

The screenshot shows the Azure DevOps Test Plans page. The left sidebar contains navigation links: Know, Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area is titled 'Test Plans' and shows a list of test plans under the 'All' tab. The list has columns for Title, Test Plan ID, State, Area Path, Iteration, and Assigned To. There are four test plans listed:

Title	Test Plan ID	State	Area Path	Iteration	Assigned To
Subject Selection Functionality	13	Active	Know	Know	Tanish
Role-based Access	10	Active	Know	Know	Tanish
Chapter-wise Course Learning Flow	6	Active	Know	Know	Tarun Aanand S G
Login verification	1	Active	Know	Know	Tarun Aanand S G

The screenshot shows the details of a test plan titled 'Chapter-wise Course Learning Flow (ID: 7)'. The left sidebar is the same as the previous screenshot. The main area shows the 'Test Suites' section with a filter by name. Below the filter, there is a table of test points:

Title	Outcome	Order	Test Case Id	Configuration	Tester
Test Case 1: Open a course and view chapters	In Progress	1	8	Windows 10	Tarun Aanand S G
Test Case 2: Complete a chapter and mark as done	Failed	2	9	Windows 10	Tarun Aanand S G

4.Installation of test

The screenshot shows the Azure DevOps Test Plans interface. On the left is a sidebar with navigation options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Runs, and Artifacts. The main content area has a header indicating the user is logged in as 'tanuamand27' and is viewing the 'Test Plans' section. Below this, there's a 'Welcome to Test hub' section with a central location for coordinating manual test activities. It includes a 'Create, run and track tests directly from the cards' section and a 'Use Test & Feedback to test web applications and submit feedback' section. A video player for 'TEST & FEEDBACK' is visible. At the bottom, there's a section for 'Upgrade to Test Manager extension to get full test management capabilities'.

The screenshot shows the Chrome Web Store page for the 'Test & Feedback' extension. The page features the extension's logo, a 'Featured' badge, a 4.2-star rating from 675 reviews, and a 'Share' button. Below this, there are two preview images: one showing the extension's interface with a 'Capture' button and another showing a 'CAPTURE' window with various settings. The page also indicates that the extension is 'Workflow & Planning' and has '200,000 users'.

5. Running the test cases

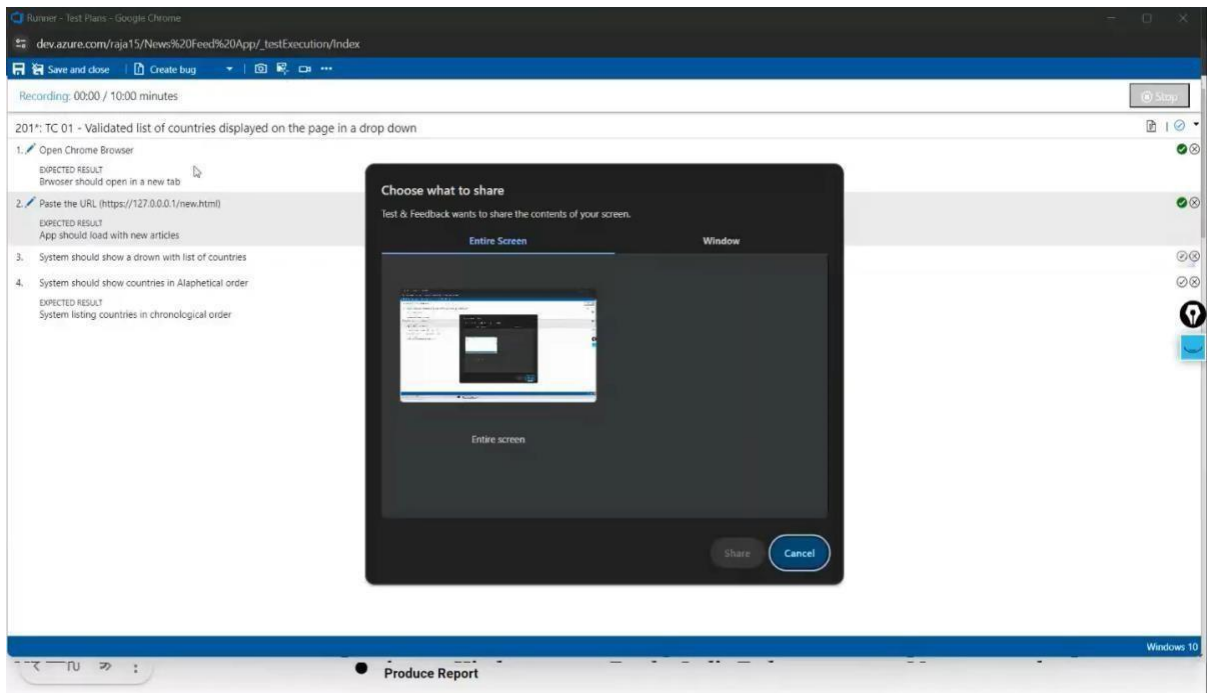
The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation options: Know, Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays the 'Subject Selection Functionality (ID: 14)' test plan. The 'Test Suites' section shows 'Subject Selection Functionality (1)'. The 'Test Points' section lists one item: 'To verify that users can successfully select a subject from the available list and that the results are displayed in sequence'. A context menu is open over the test point, showing options: View execution history, Mark Outcome, Run, Reset test to active, Edit test case, and Assign tester. The 'Run' option is selected, and a sub-menu is open showing 'Run for web application', 'Run for desktop application', and 'Run with options'.

Title	Outcome	Order	Test Case Id	Configuration	Tester
To verify that users can successfully select a subject from the available list and that the results are displayed in sequence		1	15	Windows 10	Tanish

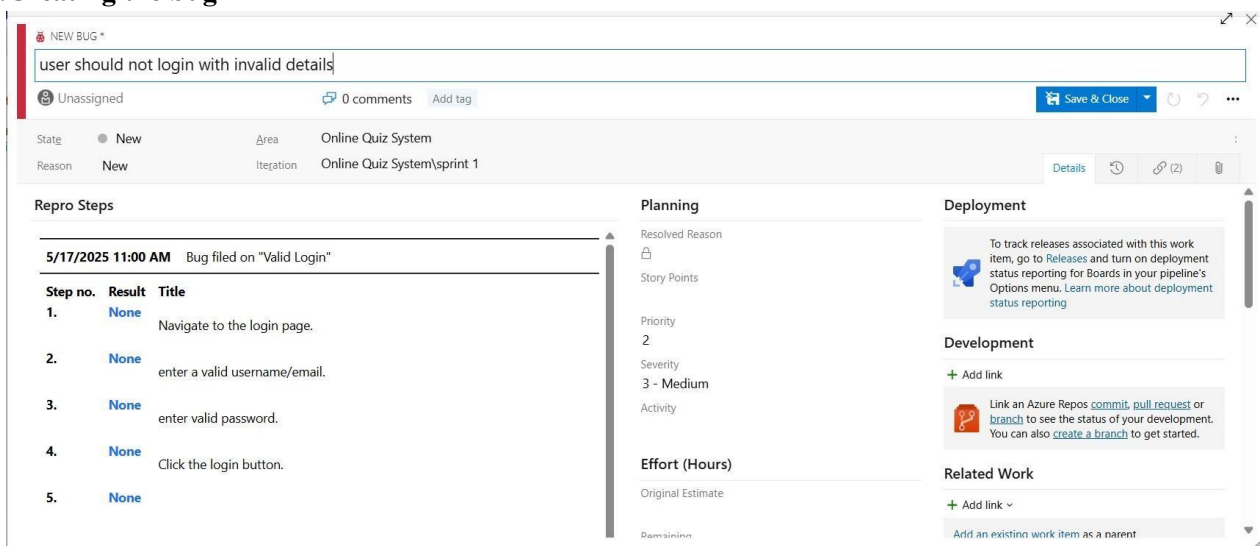
The screenshot shows the Test Runner interface in a browser. The URL is 'dev.azure.com/E-LearningPlatform-know/known/testExecution/index'. The test case is '8: Test Case 1: Open a course and view chapters'. The steps are:

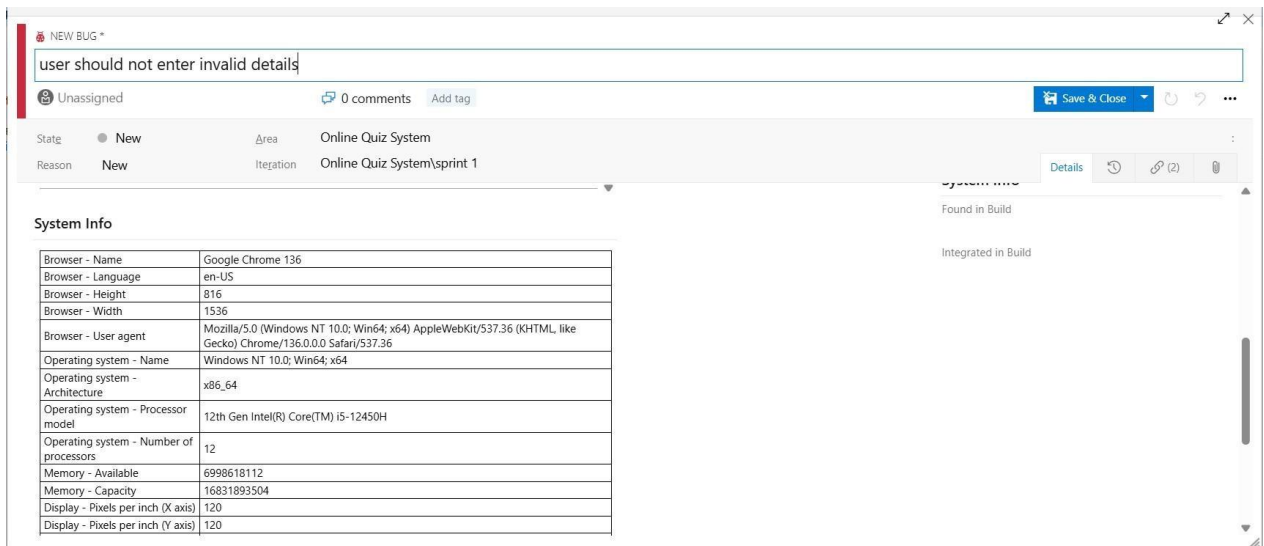
1. Login and search for a course
2. Click on a course from the result
3. Check the chapter list displayed
4. **Expected Result** | List of chapters is displayed in sequence |

6. Recording the test case

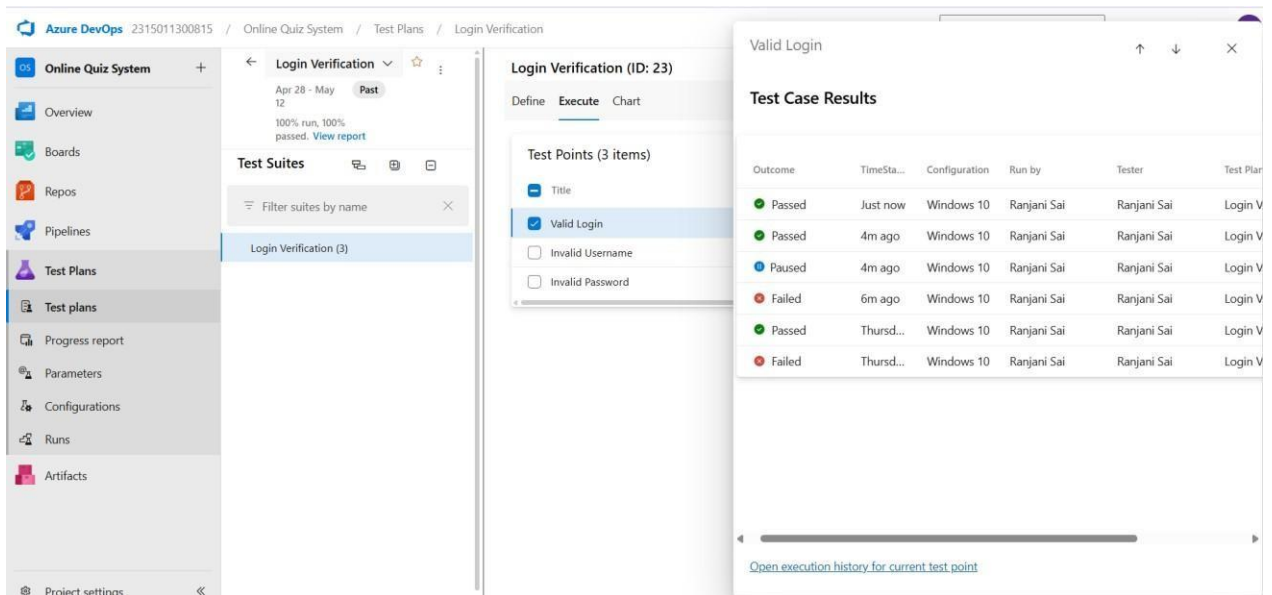


7. Creating the bug





8. Test case results



9. Test report summary

NEW BUG *

User should not enter invalid details

Unassigned

0 comments

Add tag

State

New

Area

Online Quiz System

Reason

New

Iteration

Online Quiz System\spring 1

Details

(2)

Repro Steps

5/17/2025 11:09 AM Bug filed on "Valid Login"

Step no.	Result	Title
1.	None	Navigate to the login page.
2.	None	enter a valid username/email.
3.	None	enter valid password.
4.	None	Click the login button.
5.	None	

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

From scratch

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

+ Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

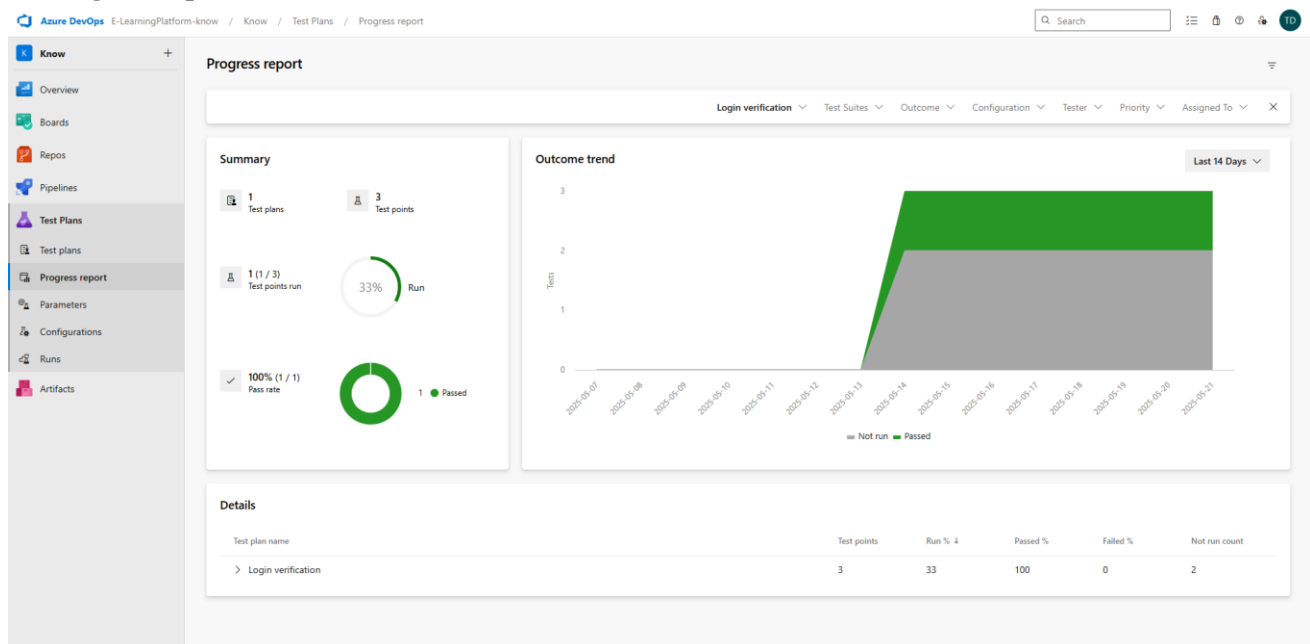
Related Work

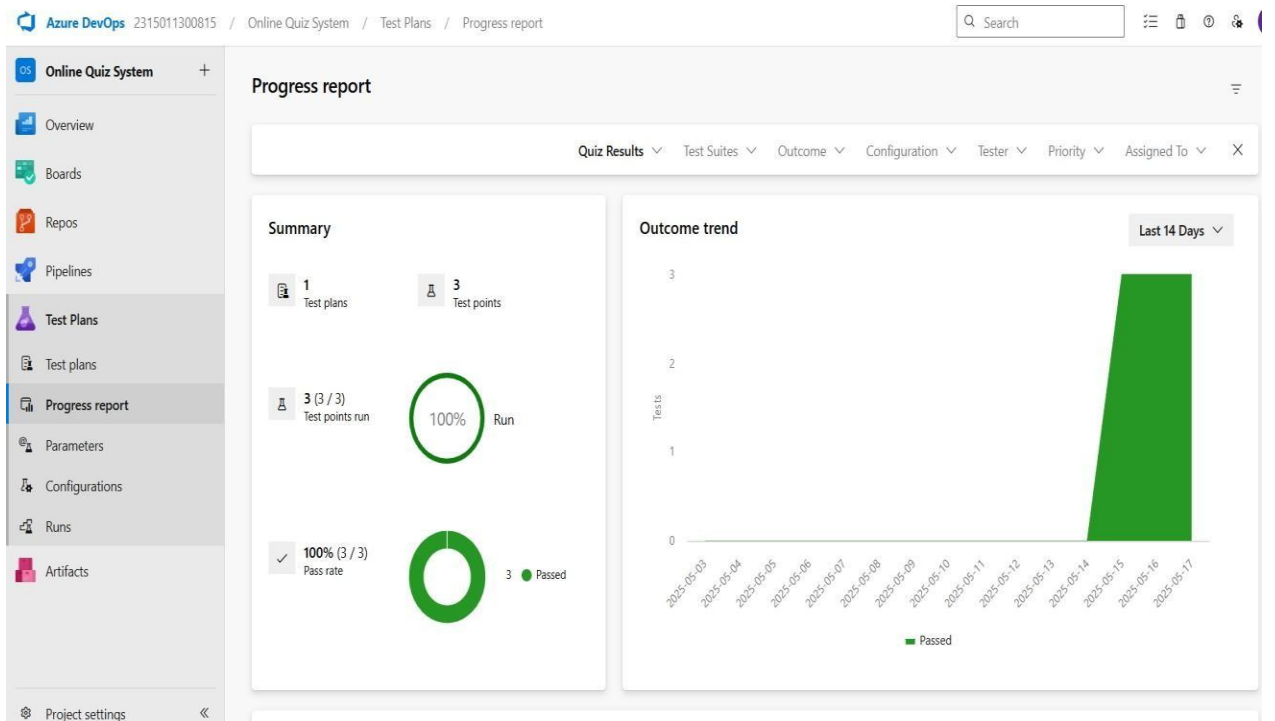
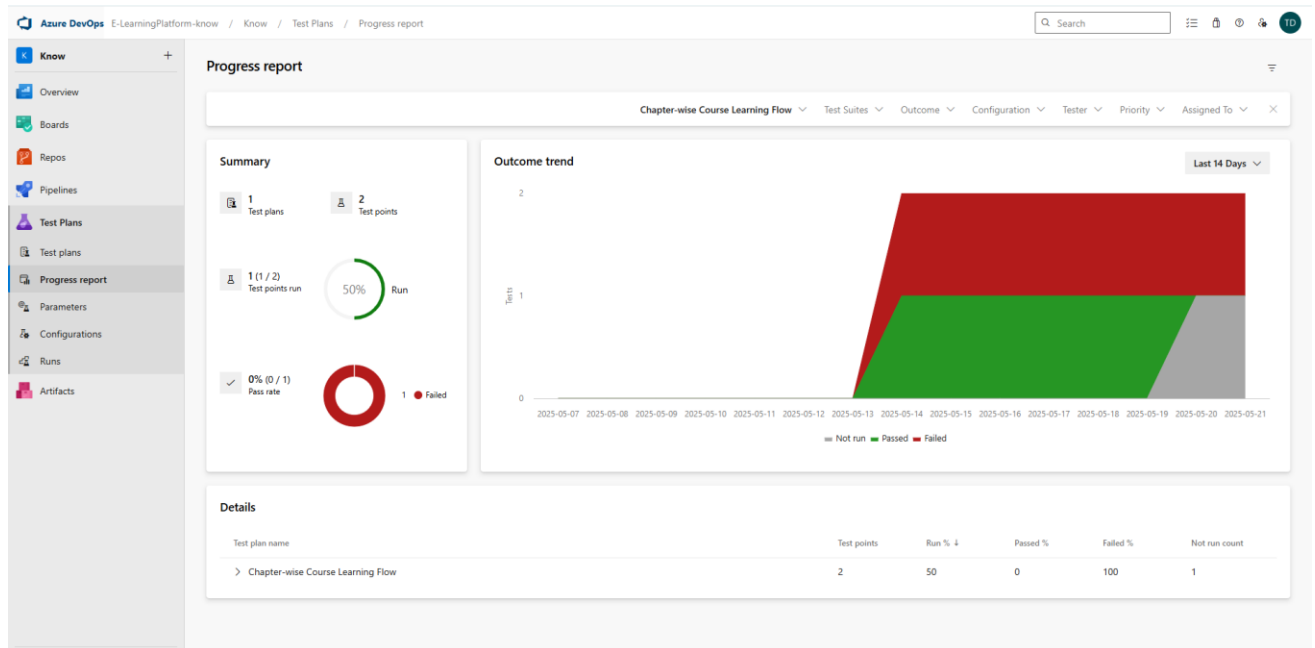
+ Add link

Add an existing work item as a parent

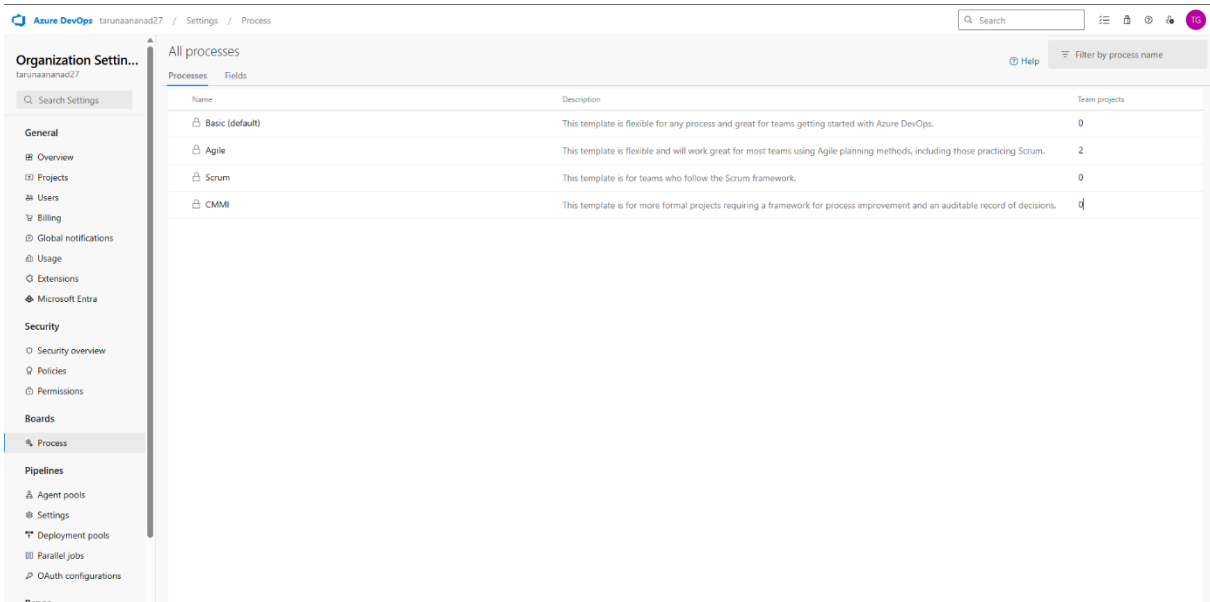
- Assigning bug to the developer and changing state

10. Progress report



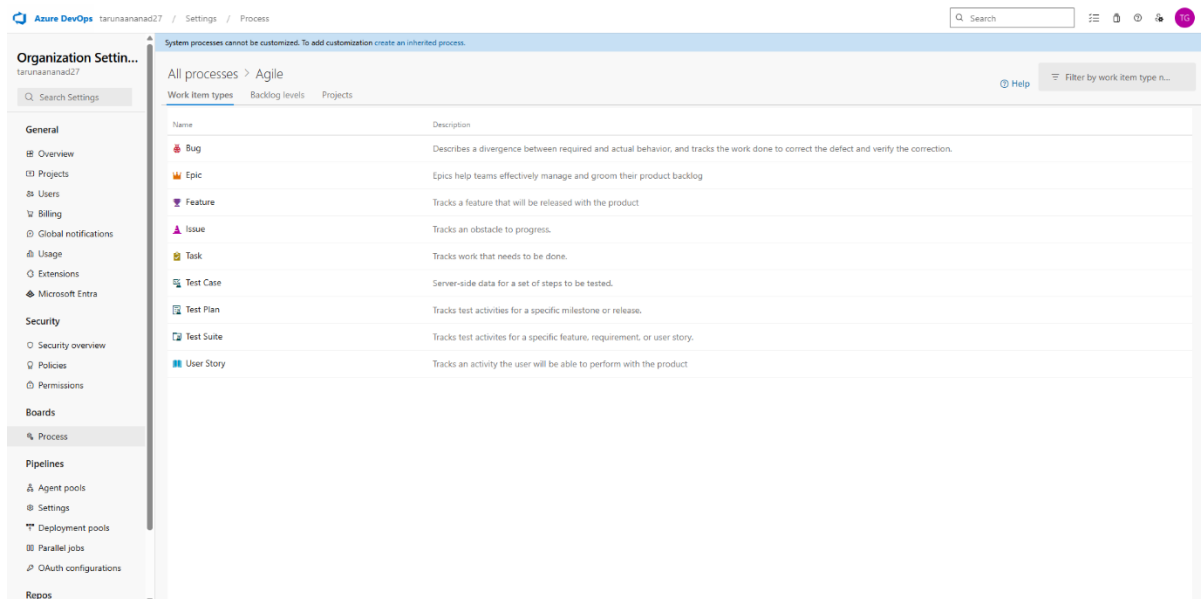


11.Changing the test template



The screenshot shows the Azure DevOps Settings page for 'tarunaananad27' under the 'Process' tab. The left sidebar contains the 'Organization Settings' menu with options like General, Security, Boards, Pipelines, and Repos. The main content area is titled 'All processes' and displays a table of available process templates.

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	2
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of decisions.	0



The screenshot shows the Azure DevOps Settings page for 'tarunaananad27' under the 'Process' tab, specifically the 'Agile' section. The left sidebar is the same as the previous screenshot. The main content area is titled 'All processes > Agile' and displays a table of work item types.

System processes cannot be customized. To add customization create an inherited process.

Name	Description
Bug	Describes a divergence between required and actual behavior, and tracks the work done to correct the defect and verify the correction.
Epic	Epics help teams effectively manage and groom their product backlog
Feature	Tracks a feature that will be released with the product
Issue	Tracks an obstacle to progress.
Task	Tracks work that needs to be done.
Test Case	Server-side data for a set of steps to be tested.
Test Plan	Tracks test activities for a specific milestone or release.
Test Suite	Tracks test activities for a specific feature, requirement, or user story.
User Story	Tracks an activity the user will be able to perform with the product

RESULT

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.

EXP NO: 9

CI/CD PIPELINES IN AZURE

AIM

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the E-Learning Platform, ensuring faster delivery and improved software quality.

PROCEDURE

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup. If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file.
6. Install dependencies (e.g., npm install, dotnet restore).

7. Build the application (dotnet build, npm run build).
8. Run unit tests (dotnet test, npm test).
9. Publish build artifacts to be used in the release stage.
10. Save and Run the Pipeline for the First Time
Save the YAML or build definition and click “Run”.
Azure will fetch the latest code and execute the defined build and test stages.
11. Configure Continuous Deployment (CD)
Navigate to the Releases tab under Pipelines and click “New Release Pipeline”. Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).
12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.
13. Set Triggers and Approvals
Enable continuous deployment trigger so the release pipeline runs automatically after a successful build. For production environments, configure pre-deployment approvals to ensure manual verification before release.
14. Monitor Pipelines and Manage Logs
View all pipeline runs under the Runs section.
Check logs for build/test/deploy stages to debug any errors.
You can also integrate email alerts or Microsoft Teams notifications for build failures.
15. Review and Maintain Pipelines
Regularly update your pipeline tasks or YAML configurations as your application grows.
Ensure pipeline runs are clean and artifacts are stored securely.
Integrate quality gates and code coverage policies to maintain code quality.

Azure DevOps E-LearningPlatform-know / Know / Pipelines

Search

Know Overview Boards Repos Pipelines Pipelines Environments Library Test Plans Artifacts

Pipelines

Recent All Runs Filter pipelines

Recently run pipelines

Pipeline	Last run
TarunAanand.e-learning (4)	#20250521.1 • Add files via upload Individual CI for 37 min 1h ago
TarunAanand.e-learning	#20250521.1 • Add files via upload Individual CI for 37 min 1h ago
Know (3)	#20250515.1 • Set up CI with Azure Pipelines PR automated for 24h Thursday 24h
Know	#20250514.4 • Update azure-pipelines.yml for Azure Pipelines Manually triggered for 7 min May 14 <1h

Azure DevOps E-LearningPlatform-know / Know / Pipelines / Know (3) / 20250515.1

Search

Know Overview Boards Repos Pipelines Pipelines Environments Library Test Plans Artifacts

#20250515.1 • Set up CI with Azure Pipelines

Know (3) Run now

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Code Coverage

Pull request by Tarun Aanand S G

Repository and version
Know
azure-pipelines 2b5ca2b4

Time started and elapsed
Thu at 8:14 PM
24s

Related
0 work items
0 artifacts

Tests and coverage
Get started

3 commits

Warnings 4

- Free disk space on C:\ is lower than 5%; Currently used: 96.15%
Checkout: Know@azure-pipelines to s
- Free disk space on C:\ is lower than 5%; Currently used: 96.15%
Run a one-line script
- Free disk space on C:\ is lower than 5%; Currently used: 96.15%
Run a multi-line script
- Free disk space on C:\ is lower than 5%; Currently used: 96.15%
Post-job: Checkout: Know@azure-pipelines to s

Jobs

Name	Status	Duration
Job	Success	16s

RESULT

Thus, the pipelines for the given project “E-Learning Platform” has been executed successfully.

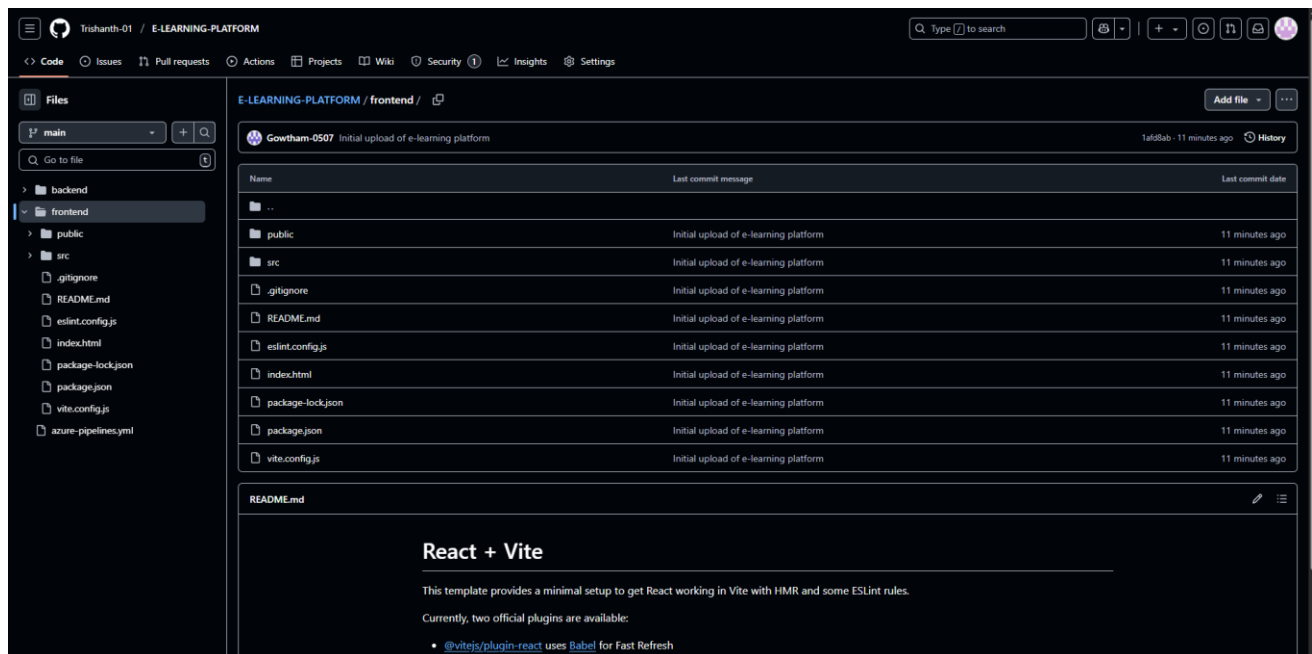
EXP NO: 10

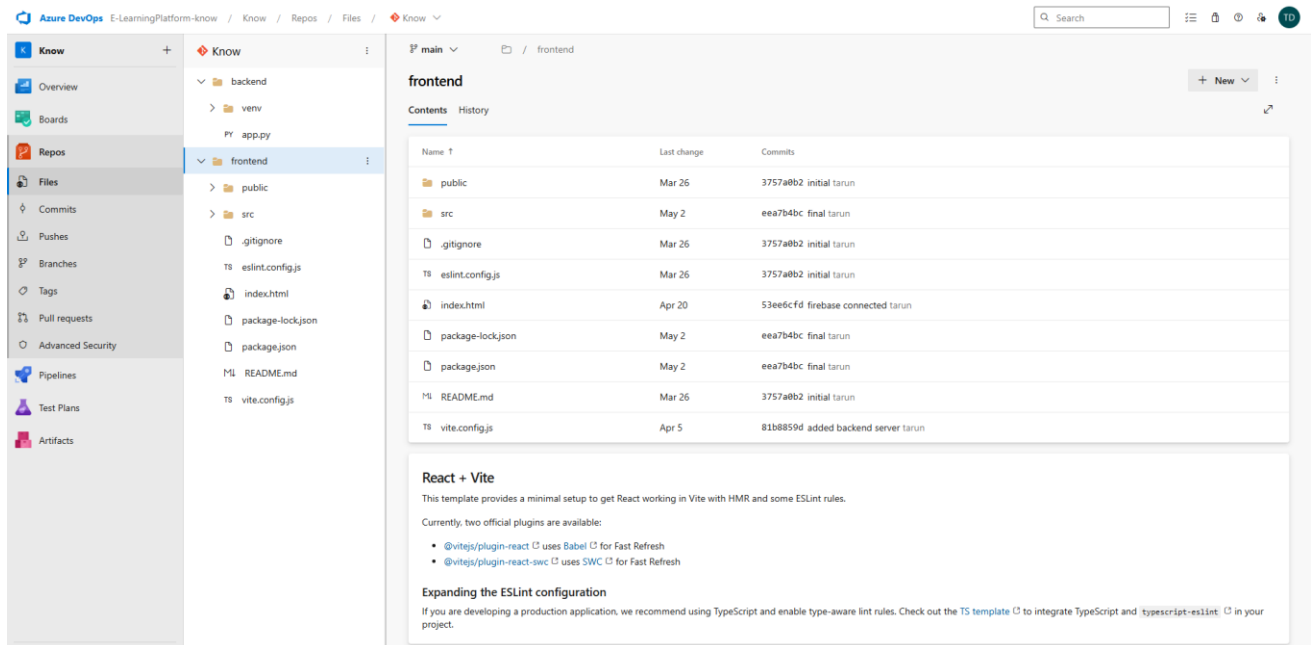
GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the E-Learning platform.

GitHub Project Structure





Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.